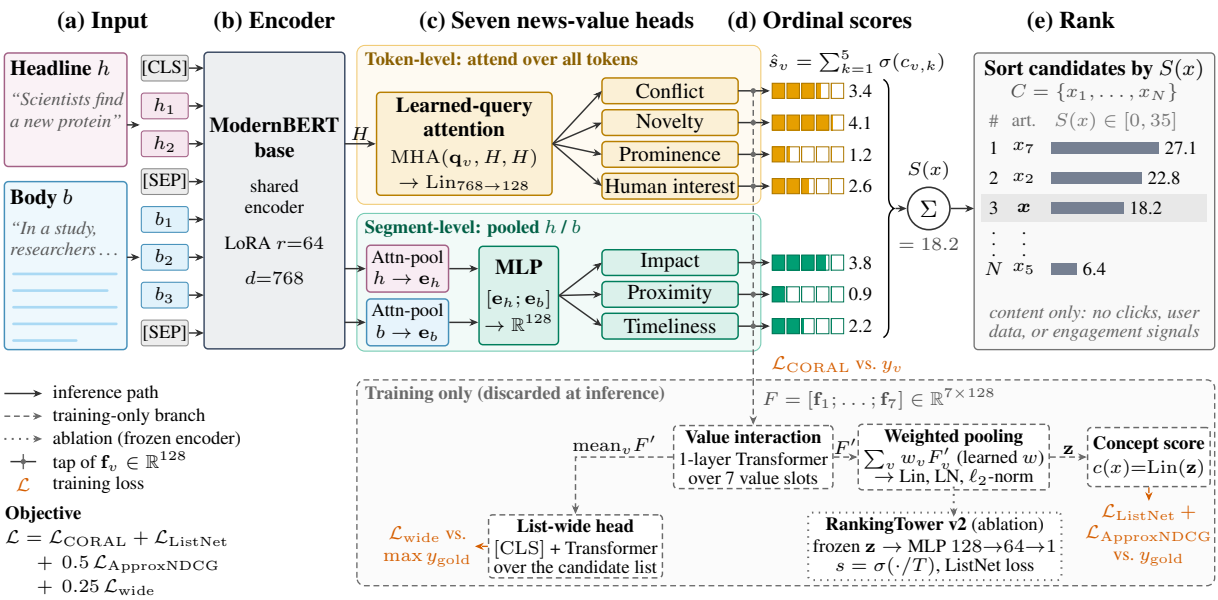

(a) Input
(b) Encoder
(c) Seven news-value heads
(d) Ordinal scores
(e) Rank
Headline h
"Scientists find a new protein"
Body b
"In a study, researchers . . ."
[CLS]
h_1
h_2
[SEP]
b_1
b_2
b_3
[SEP]
ModernBERT base
shared encoder
LoRA r=64
d=768
H
Token-level: attend over all tokens
Learned-query attention
$\mathrm{MHA}(\mathbf{q}_v, H, H)$
$\to \mathrm{Lin}_{768\to128}$
Conflict
Novelty
Prominence
Human interest
Segment-level: pooled h / b
Attn-pool $h \to \mathbf{e}_h$
Attn-pool $b \to \mathbf{e}_b$
MLP
$[\mathbf{e}_h; \mathbf{e}_b]$
$\to \mathbb{R}^{128}$
Impact
Proximity
Timeliness
$\hat{s}_v = \sum_{k=1}^{5} \sigma(c_{v,k})$
3.4
4.1
1.2
2.6
3.8
0.9
2.2
$S(x)$
Σ
$= 18.2$
Sort candidates by $S(x)$
$C = \{x_1, \ldots, x_N\}$
art. $S(x) \in [0, 35]$
1 x_7 27.1
2 x_2 22.8
3 $\boldsymbol{x}$ 18.2
N x_5 6.4
content only: no clicks, user data, or engagement signals
$\mathcal{L}_{\mathrm{CORAL}}$ vs. y_v
inference path
training-only branch
ablation (frozen encoder)
tap of $\mathbf{f}_v \in \mathbb{R}^{128}$
$\mathcal{L}$ training loss
Training only (discarded at inference)
$F = [\mathbf{f}_1; \ldots; \mathbf{f}_7] \in \mathbb{R}^{7\times128}$
$\mathrm{mean}_v F'$
Value interaction
1-layer Transformer over 7 value slots
F'
Weighted pooling
$\sum_v w_v F'_v$ (learned w)
$\to$ Lin, LN, ℓ_2-norm
$\mathbf{z}$
Concept score
$c(x) = \mathrm{Lin}(\mathbf{z})$
$\mathcal{L}_{\mathrm{ListNet}} +$
$\mathcal{L}_{\mathrm{ApproxNDCG}}$
vs. y_{gold}
$\mathcal{L}_{\mathrm{wide}}$ vs.
$\max y_{\mathrm{gold}}$
List-wide head
[CLS] + Transformer over the candidate list
RankingTower v2 (ablation)
frozen $\mathbf{z}$ $\to$ MLP 128$\to$64$\to$1
$s = \sigma(\cdot/T)$, ListNet loss
Objective
$\mathcal{L} = \mathcal{L}_{\mathrm{CORAL}} + \mathcal{L}_{\mathrm{ListNet}}$
$+ 0.5\,\mathcal{L}_{\mathrm{ApproxNDCG}}$
$+ 0.25\,\mathcal{L}_{\mathrm{wide}}$